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**SCHOOL OF
ENGINEERING**

**Bachelor of Technology
in
COMPUTER SCIENCE AND ENGINEERING**

Major Project Phase-I Report

**SMARTCROP360: An AI-Driven Conversational System for Crop
Disease diagnosis**

Batch: 119

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CERTIFICATE

This is to certify that the Major Project Stage-I work titled “**SMARTCROP360 AN AI-DRIVEN CONVERSATIONAL SYSTEM FOR CROP DISEASE DIAGNOSIS**” is carried out by **S Rakshitha (ENG22CS0423), S Ashlyn Rose (ENG22CS0421), G K Tejaswini (ENG22CS0484), Vishal A S (ENG22CS0503)**. bonafide students seventh semester of Bachelor of Technology in Computer Science and Engineering at the School of Engineering, Dayananda Sagar University, Bangalore in partial fulfillment for the award of degree in Bachelor of Technology in Computer Science and Engineering, during the year **2025-2026**.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
CLIP	Contrastive Language–Image Pretraining
CNN	Convolutional Neural Network
DL	Deep Learning
JSON	JavaScript Object Notation
LLM	Large Language Model
OCR	Optical Character Recognition
RAG	Retrieval-Augmented Generation

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ABSTRACT

Crop diseases represent a critical threat to global agricultural productivity and food security. Traditional detection methods, which rely on manual inspection and expert consultation, are slow, subjective, and inaccessible to farmers in remote areas. This project investigates the design and implementation of SmartCrop360, an AI-driven conversational system for crop disease diagnosis that integrates multimodal input processing, semantic vector retrieval, and large language model generation within a Retrieval-Augmented Generation (RAG) framework. The system analyzes both crop images and text-based symptom queries to deliver accurate disease identification and actionable treatment guidance. A CNN model was trained across 38 disease categories, achieving an overall accuracy of 96% and a macro F1-score of 0.95. CLIP-based vision–language embeddings and transformer-based sentence embeddings are projected into a shared semantic space, enabling cross-modal retrieval from a ChromaDB vector database. Retrieved agricultural context is supplied to a cloud-hosted Mistral large language model, which generates structured responses covering disease identification, symptom explanation, treatment options, and preventive recommendations. The integrated RAG pipeline significantly reduced hallucinations and improved response reliability. SmartCrop360 provides a scalable, explainable, and farmer-friendly decision-support platform suitable for modern precision agriculture.

CHAPTER 1 INTRODUCTION

Agriculture plays a fundamental role in sustaining global food supply and supporting economic development. However, crop diseases remain a major challenge, leading to reduced productivity, financial losses, and food insecurity. Early detection and proper treatment of plant diseases are essential to ensure healthy crop growth and sustainable farming practices. Traditional disease identification methods rely on manual inspection and expert consultation, which can be time-consuming, subjective, and inaccessible to farmers in remote areas.

Recent advancements in artificial intelligence and deep learning have enabled automated crop disease detection using image-based classification models. While these systems improve diagnostic accuracy, many existing solutions focus solely on predicting disease labels and do not provide detailed explanations or actionable recommendations. Additionally, limited integration of domain knowledge restricts their ability to generate context-aware responses.

To address these limitations, this project proposes SmartCrop360, an intelligent agriculture-support system that integrates multimodal input processing, semantic retrieval, and large language models within a Retrieval-Augmented Generation (RAG) framework. By combining CLIP-based image understanding, transformer-based text embeddings, and a cloud-hosted Mistral language model, the system delivers accurate disease identification along with meaningful treatment and prevention guidance.

1.1. BACKGROUND AND MOTIVATION

Crop diseases significantly reduce agricultural productivity and cause economic losses worldwide. Traditional diagnosis methods rely on manual inspection or expert consultation, which are slow, subjective, and inaccessible in rural areas. Although AI-based image classification models have improved disease detection accuracy, most existing systems only provide disease labels without detailed explanations or treatment guidance.

The motivation behind SmartCrop360 is to develop an intelligent, multimodal system that not only detects crop diseases from images and text inputs but also retrieves relevant agricultural knowledge and generates context-aware recommendations. By integrating vision models, semantic retrieval, and large language models within a RAG framework, the system aims to provide accurate, scalable, and practical advisory support for modern precision agriculture.

The proposed system also promotes sustainable agriculture by reducing unnecessary pesticide usage through precise recommendations. Its modular architecture makes it adaptable for different crops, regions, and languages, increasing accessibility for farmers in rural and resource-constrained environments.

1.2. SCOPE

The scope of this project is to design and develop an intelligent, AI-driven crop disease detection and advisory system capable of supporting farmers through multimodal interaction. The system integrates image-based disease analysis and text-based symptom queries within a unified framework to provide accurate, real-time recommendations.

The project includes the development of a CLIP-based vision–language model for extracting semantic features from crop images, transformer-based embeddings for text processing, and a Retrieval-Augmented Generation (RAG) pipeline for knowledge retrieval. A vector database (ChromaDB) is used to store and retrieve agricultural knowledge efficiently. The system integrates a cloud-hosted Mistral large language model to generate context-aware disease identification, explanations, and treatment suggestions.

The scope also covers backend development using FastAPI, semantic search implementation, system logging, and performance evaluation. The framework is designed to be modular, scalable, and extensible, allowing future integration of additional crops, diseases, multilingual support, and mobile deployment. The current

implementation focuses primarily on disease identification and advisory support rather than automated pesticide application or hardware-based field monitoring.

CHAPTER 2 PROBLEM DEFINITION

Agriculture plays a vital role in ensuring food security and economic stability, yet crop diseases continue to cause significant yield losses worldwide. Farmers often rely on manual inspection, personal experience, or delayed expert consultation to identify plant diseases. These traditional methods are time-consuming, subjective, and often inaccurate, especially in rural or resource-constrained environments. Early-stage symptoms can be difficult to detect, and misdiagnosis frequently leads to improper pesticide usage, increased production costs, and environmental harm.

Existing technological solutions typically focus on standalone image classification models that predict disease labels based solely on visual input. However, such systems often lack contextual reasoning, explanation capabilities, and integration with domain-specific agricultural knowledge. Moreover, many systems do not support multimodal interaction, limiting users to either image-only or text-only inputs. They also fail to provide comprehensive recommendations that include causes, preventive measures, and treatment strategies.

There is a clear need for an intelligent, scalable, and multimodal agricultural advisory system that can process both crop images and textual symptom descriptions, retrieve relevant knowledge from a structured database, and generate context-aware, actionable recommendations. The problem addressed in this work is to design and implement an AI-driven framework that integrates vision–language models, semantic retrieval mechanisms, and large language models to deliver accurate, explainable, and real-time crop disease identification and treatment guidance.

CHAPTER 3 LITERATURE SURVEY

A review of existing literature reveals rapid advancements in AI-based agricultural disease detection and intelligent advisory systems. Key themes emerging from the survey are summarised below.

OVERVIEW OF AI IN AGRICULTURE

Early agricultural AI solutions relied on rule-based systems and traditional machine learning techniques that required manual feature engineering. With the rise of deep learning, especially Convolutional Neural Networks (CNNs), agricultural image analysis improved significantly, enabling automatic feature extraction and higher classification accuracy. However, most systems focused only on classification outputs without providing contextual explanations or treatment recommendations [1].

PLANT DISEASE DETECTION USING DEEP LEARNING

Traditional image-processing techniques such as colour thresholding and handcrafted feature extraction achieved moderate success under controlled conditions but failed under real-world variability. CNN-based models enabled automated hierarchical visual feature learning. Despite strong classification performance, CNN-based systems remain primarily discriminative and typically provide only disease labels without explanations or preventive strategies, limiting their practical value [4].

VISION–LANGUAGE MODELS AND MULTIMODAL EMBEDDINGS

Recent developments in multimodal AI introduced vision–language models capable of learning shared representations for images and text. Contrastive learning approaches such as CLIP allow models to map visual and textual information into the same

semantic embedding space. Unlike OCR-based approaches, vision-language embeddings directly capture meaningful visual patterns such as discoloration, texture irregularities, and structural changes, enabling semantic matching without handcrafted labels [3][4].

RETRIEVAL-AUGMENTED GENERATION (RAG) AND VECTOR DATABASES

Large Language Models offer strong natural language generation capabilities but may produce inaccurate responses without external grounding. Retrieval-Augmented Generation addresses this by retrieving relevant domain-specific information before response generation. Agricultural knowledge is converted into vector embeddings and stored in semantic vector databases such as ChromaDB. Top-k similarity search retrieves relevant entries, providing the LLM with grounded context and reducing hallucinations [5][8].

LARGE LANGUAGE MODELS FOR AGRICULTURAL ADVISORY

LLMs have demonstrated strong capability in generating human-like explanations and structured responses. In agricultural applications, LLMs synthesise retrieved information into understandable recommendations. Combining LLMs with retrieval-based grounding ensures factual accuracy and allows the model to provide detailed explanations, treatment strategies, and preventive measures tailored to user queries [7][10].

GAP AND MOTIVATION

Despite advancements in agricultural AI, several limitations remain. Many systems focus exclusively on classification accuracy without contextual reasoning. Recommendation-based approaches often rely on static rule-based logic. There remains

a need for a unified multimodal system combining visual recognition, semantic retrieval, and grounded language generation. SmartCrop360 addresses this gap by integrating deep learning classification, multimodal embeddings, retrieval-based grounding, and LLM-driven advisory generation [1][5][7].

CHAPTER 4 PROJECT DESCRIPTION

SmartCrop360 is an AI-powered agricultural platform designed to assist farmers in accurately identifying crop diseases and receiving structured, context-aware treatment recommendations. The system supports both image-based and text-based inputs and delivers explainable, grounded responses through a Retrieval-Augmented Generation pipeline.

4.1. PROPOSED DESIGN

When a crop image is uploaded, it is first processed by a trained CNN model to predict the disease class. Simultaneously, the image is encoded using a CLIP-based vision–language model to generate semantic embeddings that capture visual features such as lesion distribution, colour variations, and texture abnormalities.

For text-based queries, the input is normalised and encoded using Sentence Transformer embeddings. Both image and text embeddings are projected into a shared semantic space, enabling unified multimodal retrieval from ChromaDB. The vector database stores curated agricultural knowledge including disease descriptions, symptoms, treatment methods, soil conditions, and preventive strategies.

When a user submits a query, similarity search retrieves the most relevant knowledge entries. These are aggregated into a contextual prompt and sent to the cloud-hosted Mistral LLM, which generates a structured JSON response including disease identification, symptom explanation, chemical and organic treatment options, and preventive recommendations.

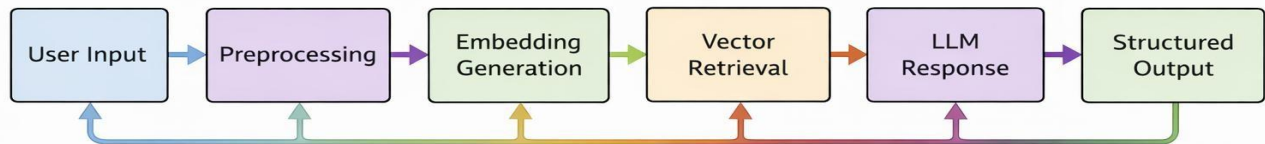


Fig 4.1: System Architecture Diagram

4.2. ASSUMPTIONS AND DEPENDENCIES

The following assumptions and dependencies underpin the SmartCrop360 system:

Assumptions

- Users have access to a smartphone or computer with a camera for uploading crop images.
- Internet connectivity is available for accessing the cloud-hosted Mistral LLM.
- The uploaded crop images are of sufficient quality and resolution for feature extraction.
- The agricultural knowledge base stored in ChromaDB is curated, verified, and regularly maintained.
- Users may input either an image or a textual symptom description, or both.

Dependencies

- CLIP model (OpenAI) — for vision–language embedding generation.

- Sentence Transformers / Nomic Embeddings — for text-based semantic encoding.
- ChromaDB — vector database for embedding storage and top-k semantic retrieval.
- Mistral LLM (cloud-hosted) — for context-aware response generation.
- FastAPI and Uvicorn — for backend REST API development and deployment.
- PyTorch and HuggingFace Transformers — for model loading and inference.

CHAPTER 5 REQUIREMENTS

5.1. FUNCTIONAL REQUIREMENTS

The SmartCrop360 system must fulfil the following functional requirements:

Image-Based Disease Detection

- The system shall accept crop images (leaf, stem, or fruit) as input.
- The system shall preprocess the image (resize, normalise) and generate CLIP-based semantic embeddings.
- The system shall classify the disease using a trained CNN model.

Text-Based Query Processing

- The system shall accept natural language symptom descriptions as input.
- The system shall convert text inputs into dense semantic vectors using Sentence Transformers or Nomic embeddings.

Knowledge Retrieval

- The system shall perform top-k semantic similarity search in ChromaDB.
- The system shall retrieve the most contextually relevant agricultural knowledge entries.

Response Generation

- The system shall pass retrieved knowledge as context to the Mistral LLM.
- The system shall generate structured JSON responses covering disease identification, symptoms, treatment, and prevention.

API Endpoints

Table 5.1: API Endpoints of the SmartCrop360 System

Method	Endpoint	Description
POST	/upload	Upload crop image → CLIP embedding → Retrieval → Mistral response
POST	/ask	Text query → Embedding → Retrieval → LLM response
GET	/retrieve	Returns raw top-k similarity results from ChromaDB
GET	/health	Backend health check

5.2. NON-FUNCTIONAL REQUIREMENTS

Software Requirements

The system relies on the following software stack:

- FastAPI — REST API development framework
- Uvicorn — ASGI server for deployment
- Pydantic — data validation and schema management
- CLIP (OpenAI / HuggingFace) — vision–language embedding model
- Sentence Transformers / Nomic Embeddings — text embedding models
- ChromaDB — vector database for embedding storage and retrieval
- Mistral LLM (cloud-hosted) — language model for response generation
- PyTorch, NumPy, Pandas, Pillow, OpenCV — supporting ML and data libraries

Hardware Requirements

Table 5.2: Hardware Requirements

Configuration	Specification
Minimum RAM	8 GB
Recommended RAM	16 GB

Configuration	Specification
Processor	Intel i5 / Ryzen 5 or higher
Storage	10 GB free (SSD recommended)
GPU	NVIDIA GPU (optional, for faster embedding generation)
Internet	Required for cloud LLM access

CHAPTER 6 IMPLEMENTATION APPROACH

The SmartCrop360 system is implemented as a modular, multi-stage pipeline. Each stage is independently verifiable and extensible, ensuring scalability and ease of maintenance.

6.1. IMPLEMENTATION STRATEGY

Stage 1: Input Stage

The FastAPI backend receives user input either as a crop image upload or a text query. Images are resized and normalised before being forwarded to the CLIP model for semantic embedding generation. Text inputs are cleaned, normalised, and encoded using Sentence Transformer or Nomic models. The backend routes each input type to its corresponding processing pipeline.

Stage 2: Multimodal Embedding Generation

SmartCrop360 employs a unified semantic embedding space. CLIP captures visual disease characteristics such as lesion distribution, colour variation, and texture changes from crop images. Sentence Transformer or Nomic embeddings encode semantic meaning from textual symptom descriptions. Both modalities are projected into a shared vector space, enabling cross-modal retrieval between image and text data.

Stage 3: Vector Storage and Retrieval

Agricultural knowledge entries — including crop name, disease name, symptoms, causes, chemical treatments, organic remedies, and preventive measures — are pre-encoded and stored in ChromaDB with associated metadata. When a query is submitted, the system converts the input into an embedding and performs top-k similarity search. The most semantically relevant knowledge chunks are retrieved to form the basis of the LLM prompt.

Stage 4: Retrieval-Augmented Generation

Retrieved knowledge entries are structured into a contextual prompt and passed to the cloud-hosted Mistral LLM. The LLM generates a grounded response covering disease identification, symptom explanation, treatment recommendations, and preventive guidance. Since the model receives verified agricultural context, hallucinations are minimised and factual accuracy is improved. The final output is formatted as structured JSON for seamless integration with web or mobile frontends.

Stage 5: Logging and Monitoring

The system optionally logs user queries, retrieved documents, and generated responses. These logs support continuous performance monitoring, knowledge base expansion, and system improvement over time. Performance evaluation was conducted using Kaggle, enabling rapid benchmarking of CNN classification, CLIP embedding quality, and retrieval accuracy.

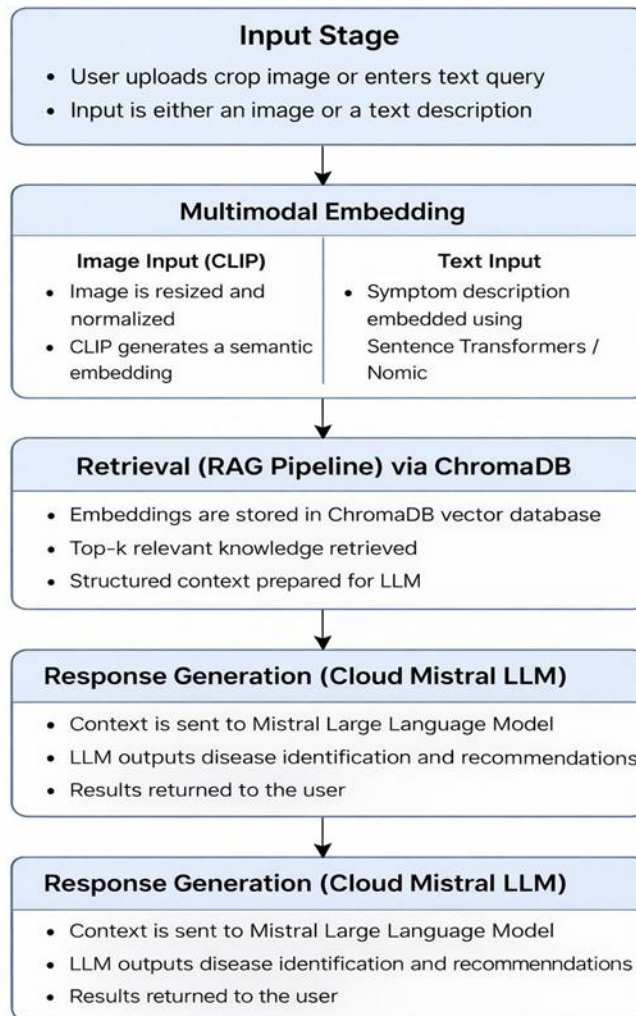


Fig 6.1: Overall Workflow of the SmartCrop360 System

CHAPTER 7 EXPERIMENTATION

The experimentation phase focused on evaluating the performance and stability of the CNN-based crop disease classification model and the integrated RAG pipeline.

7.1. DATASET AND TRAINING SETUP

The dataset consisted of 38 disease categories covering multiple crops, split into training and validation subsets. The CNN model was trained for 5 epochs using the Adam optimiser with appropriate learning rate scheduling. Data augmentation techniques including rotation, flipping, and brightness adjustment were applied to improve generalisation.

7.2. CNN MODEL TRAINING RESULTS

During training, both training and validation metrics were monitored to assess convergence behaviour. Validation accuracy steadily improved across epochs, reaching a peak of 96.32% at Epoch 4, after which a slight decline was observed, indicating mild overfitting. Therefore, the best-performing model checkpoint was retained. Training and validation loss curves demonstrated stable convergence without significant oscillations, confirming effective learning.

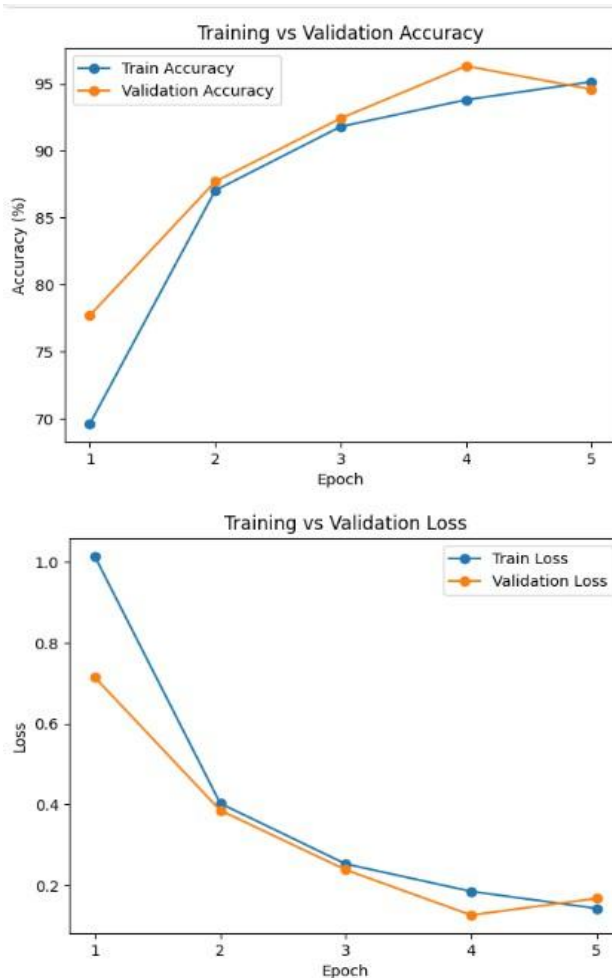


Fig 7.2: Training vs Validation Accuracy and Loss Graph

7.3. RAG PIPELINE EVALUATION

Experimentation on the RAG component involved evaluating top-k similarity retrieval performance using ChromaDB. Top-3 retrieval was found to provide optimal contextual grounding for the LLM, reducing irrelevant responses. Structured metadata inclusion further minimised hallucination and improved response consistency. Prompt experimentation with Mistral confirmed that including structured retrieved context significantly improved the quality and factual accuracy of generated responses.

CHAPTER 8 RESULT AND DISCUSSION

The proposed SmartCrop360 framework achieved strong performance across all evaluated components, demonstrating the effectiveness of integrating CNN classification, multimodal embeddings, semantic retrieval, and LLM-based response generation.

8.1. CLASSIFICATION PERFORMANCE

The final CNN model attained an overall accuracy of 96%, with a macro F1-score of 0.95 and a weighted F1-score of 0.96 across 38 crop disease categories. Precision and recall remained consistently high across most categories, demonstrating robust generalisation capability. The confusion matrix exhibited strong diagonal dominance, indicating minimal inter-class misclassification. Minor confusion was observed between visually similar diseases such as Early Blight and Late Blight.

Classification Report				
	precision	recall	f1-score	support
Apple___Apple_scab	0.94	0.93	0.94	73
Apple___Black_rot	0.94	0.96	0.95	53
Apple___Cedar_apple_rust	0.96	1.00	0.98	24
Apple___healthy	0.94	0.87	0.90	156
Background_without_leaves	0.99	0.98	0.99	112
Blueberry___healthy	0.97	0.97	0.97	154
Cherry___Powdery_mildew	0.96	1.00	0.98	114
Cherry___healthy	1.00	0.98	0.99	83
Corn___Cercospora_leaf_spot Gray_leaf_spot	0.98	0.82	0.89	65
Corn___Common_rust	0.98	1.00	0.99	130
Corn___Northern_Leaf_Blight	0.86	0.97	0.91	92
Corn___healthy	1.00	0.99	1.00	124
Grape___Black_rot	0.94	0.96	0.95	122
Grape___Esca_(Black_Measles)	0.95	0.98	0.97	128
Grape___Leaf_blight_(Isariopsis_Leaf_Spot)	0.96	0.97	0.97	113
Grape___healthy	0.97	1.00	0.99	38
Orange___Haunglongbing_(Citrus_greening)	1.00	0.99	1.00	526
Peach___Bacterial_spot	0.98	0.99	0.98	240
Peach___healthy	1.00	0.94	0.97	34
Pepper,_bell___Bacterial_spot	0.95	0.97	0.96	105
Pepper,_bell___healthy	0.96	0.95	0.96	155
Potato___Early_blight	1.00	0.95	0.97	92
Potato___Late_blight	0.83	0.97	0.89	87
Potato___healthy	0.80	1.00	0.89	12
Raspberry___healthy	1.00	0.64	0.78	42
Soybean___healthy	0.97	0.99	0.98	541
Squash___Powdery_mildew	1.00	0.96	0.98	186
Strawberry___Leaf_scorch	1.00	0.96	0.98	98
Strawberry___healthy	1.00	0.96	0.98	45
Tomato___Bacterial_spot	0.98	0.94	0.96	179
Tomato___Early_blight	0.84	0.85	0.85	107
Tomato___Late_blight	0.92	0.94	0.93	186
Tomato___Leaf_Mold	0.97	0.96	0.97	101
Tomato___Septoria_leaf_spot	1.00	0.87	0.93	158
Tomato___Spider_mites Two-spotted_spider_mite	0.99	0.90	0.94	164
Tomato___Target_Spot	0.82	0.94	0.88	126
Tomato___Tomato_Yellow_Leaf_Curl_Virus	0.97	1.00	0.98	575
Tomato___Tomato_mosaic_virus	0.91	1.00	0.95	42
Tomato___healthy	0.99	1.00	0.99	164
accuracy			0.96	5546
macro avg	0.96	0.95	0.95	5546
weighted avg	0.96	0.96	0.96	5546

Fig. 8.1: Classification Report Table (Precision, Recall, F1-score)

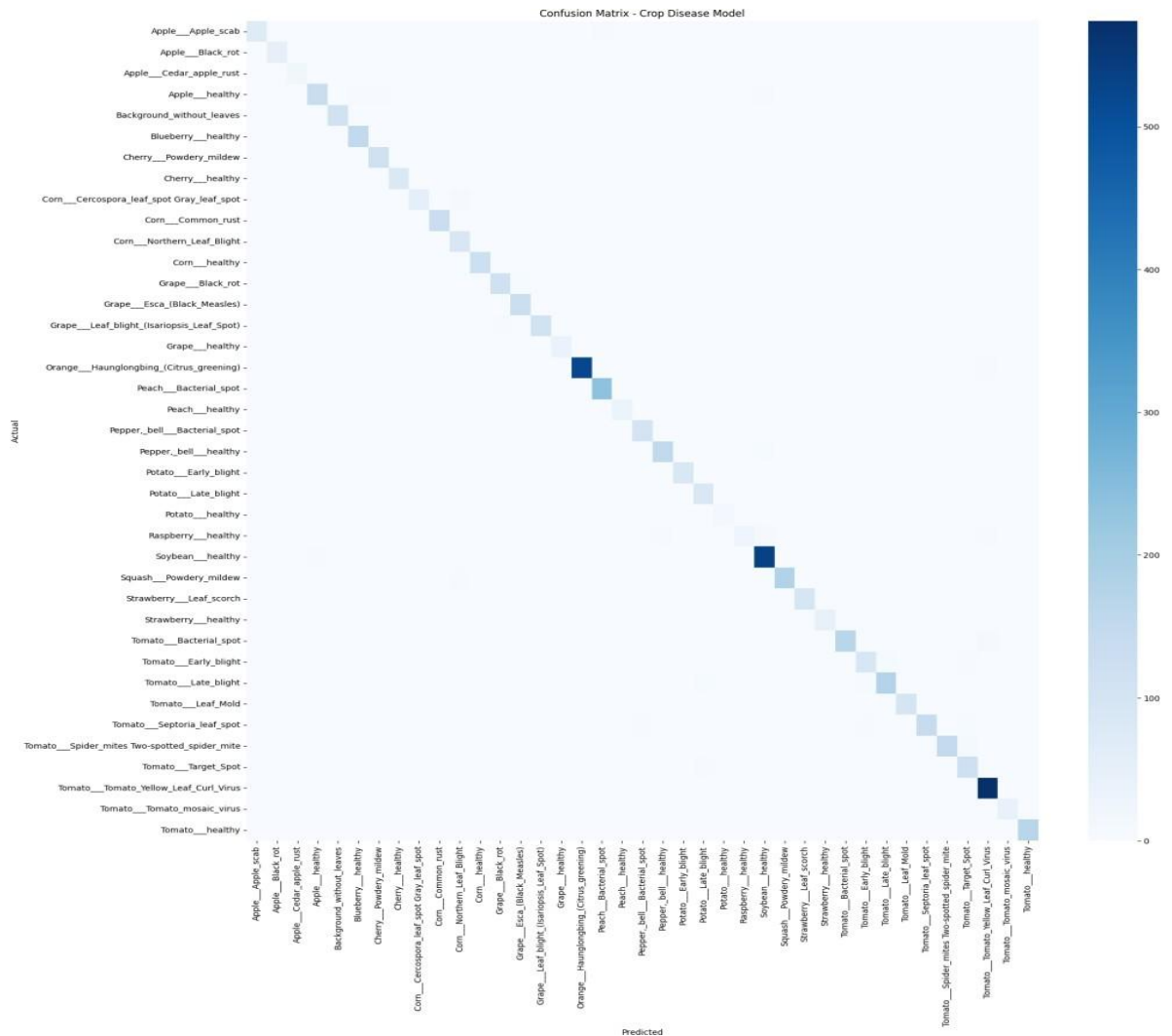


Fig. 8.2: Confusion Matrix

8.2. SYSTEM INTERFACE AND ADVISORY OUTPUT

The SmartCrop360 web interface provides farmers with an intuitive platform for disease detection and advisory guidance. Users can upload crop images or enter symptom descriptions, and the system returns structured recommendations covering disease identification, severity level, treatment options, and preventive strategies. The AI chatbot interface enables conversational interaction, answering crop-related questions in natural language.

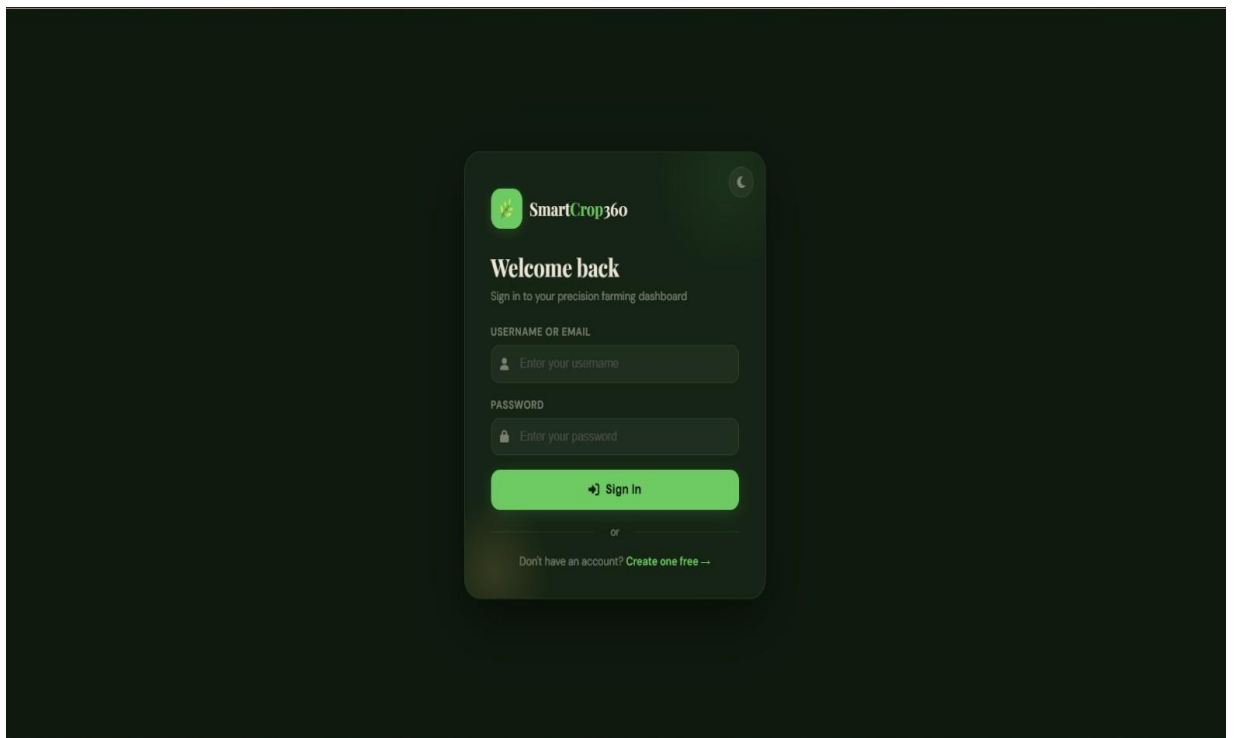


Fig. 8.3: Login Page

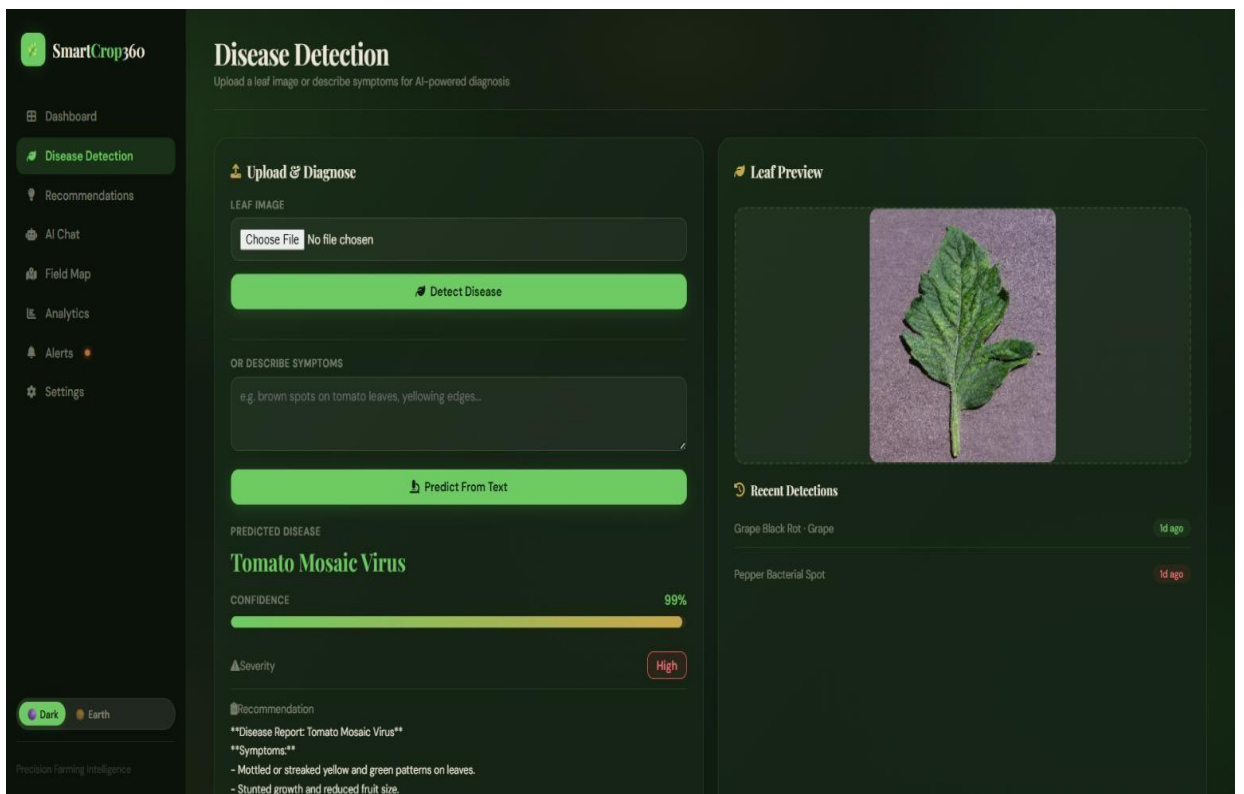


Fig. 8.4: Disease Detection Page

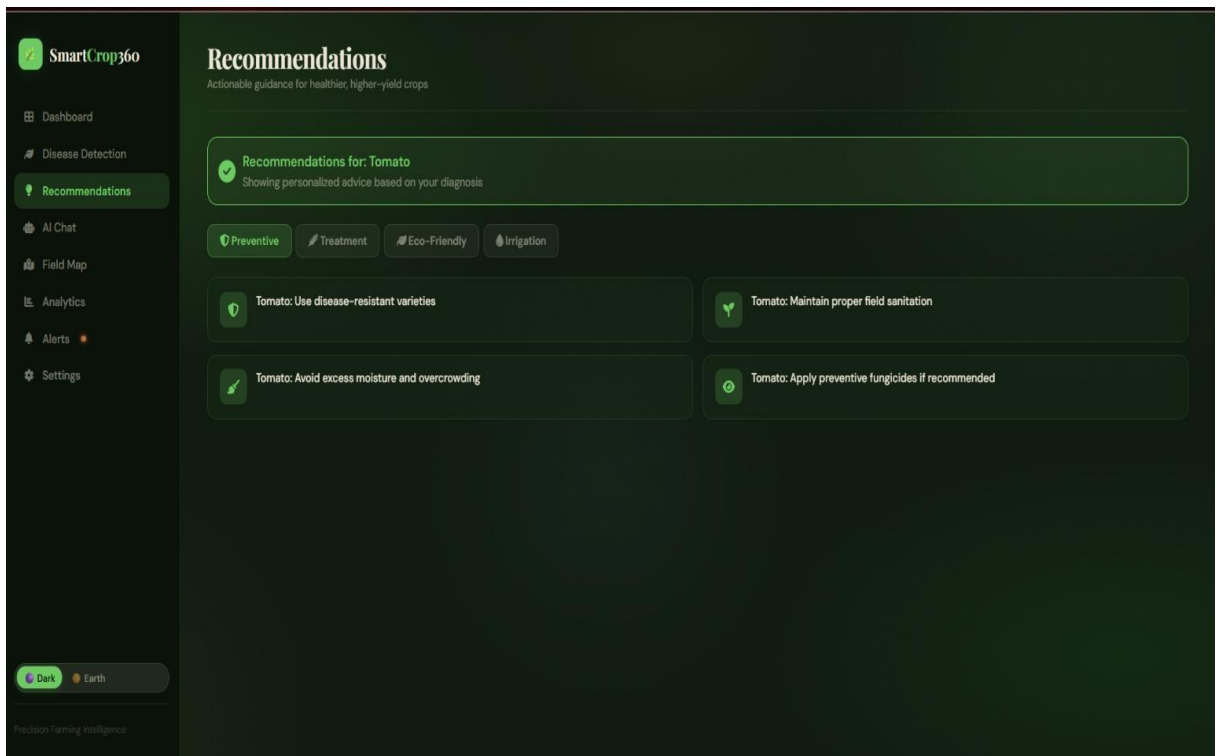


Fig. 8.5: Recommendations (Prevention)

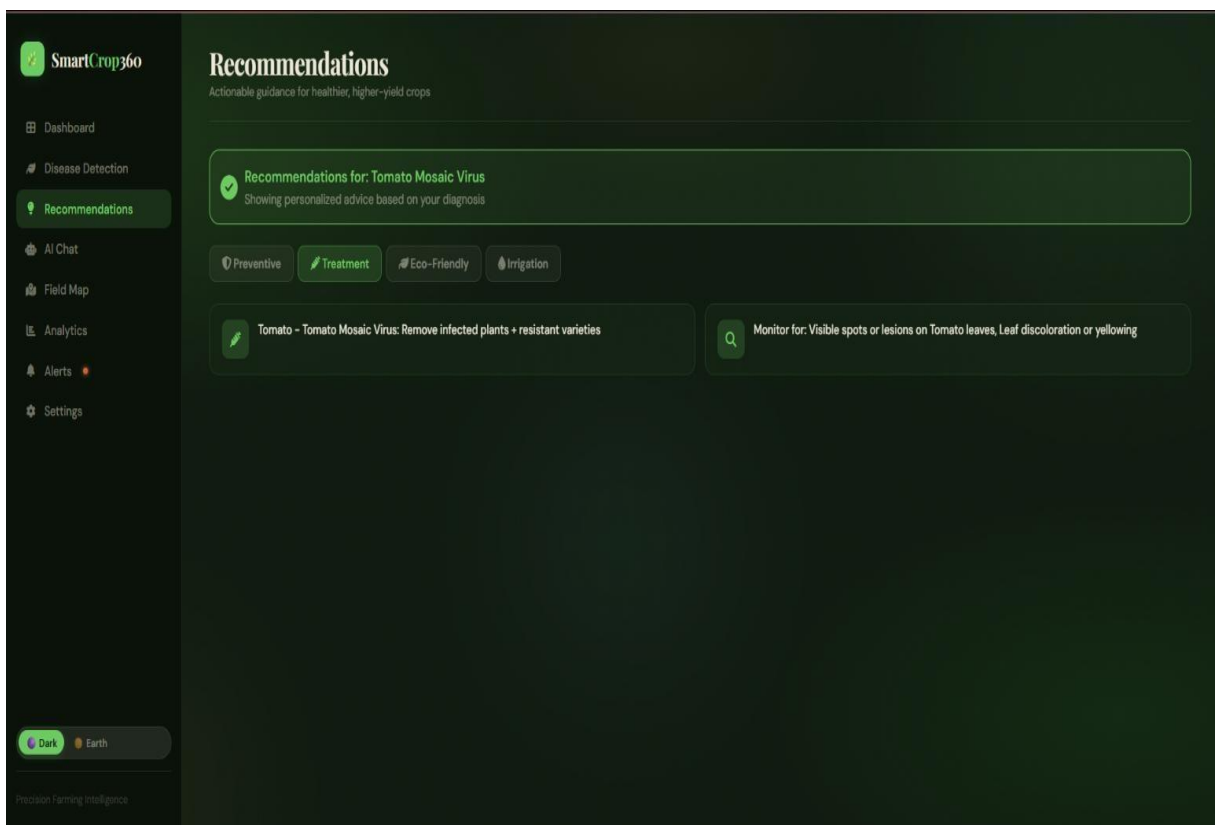


Fig. 8.6: Recommendations (Treatment)

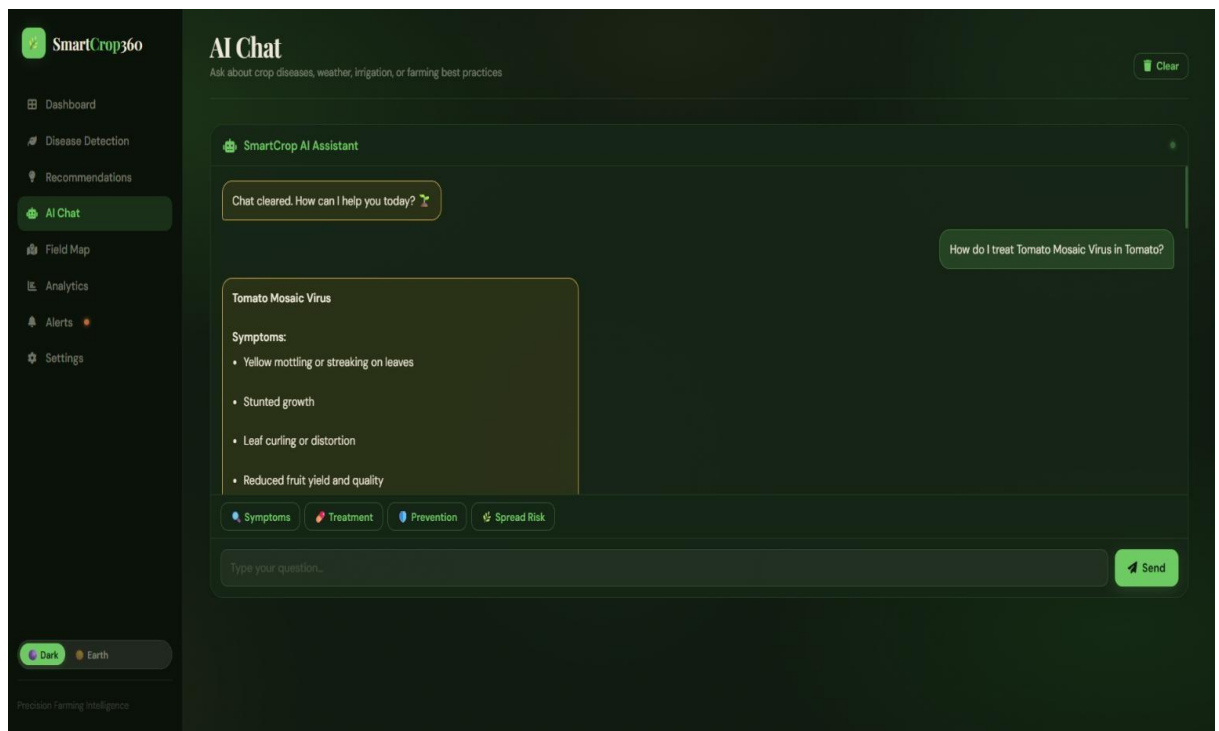


Fig. 8.7: AI Chatbot Interface

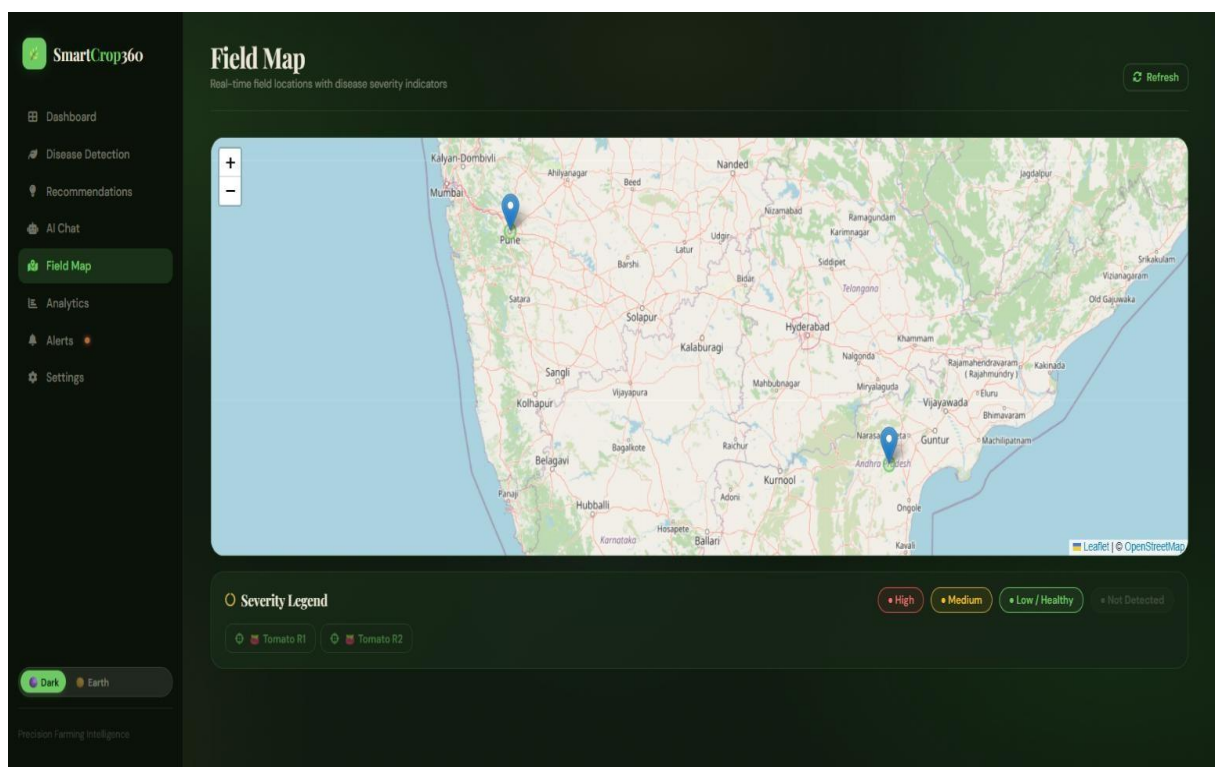


Fig. 8.8: Field Map

8.3. DISCUSSION

The integration of CNN classification with retrieval-grounded LLM generation ensured both predictive accuracy and contextual reliability. Grounding the Mistral LLM with top-3 semantically retrieved documents significantly improved response quality and reduced hallucinations. The structured JSON output enables seamless integration with web or mobile agricultural advisory platforms. Overall, SmartCrop360 successfully bridges the gap between advanced AI technologies and practical farming applications, offering an explainable and scalable solution for precision agriculture.

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